INCIDENT LIFE CYCLE AUTOMATION

# Functional System Design (FSD) Documentation

Implementation-oriented design for Incident Life Cycle Automation

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| STUDENT NAME<br>**Shanmugapriya M** | PROJECT<br>**Incident Life Cycle Automation** |
| DOCUMENT DATE<br>**08 October 2026** | SCOPE<br>**Incident lifecycle workflow & automation only** |

## 1. Introduction

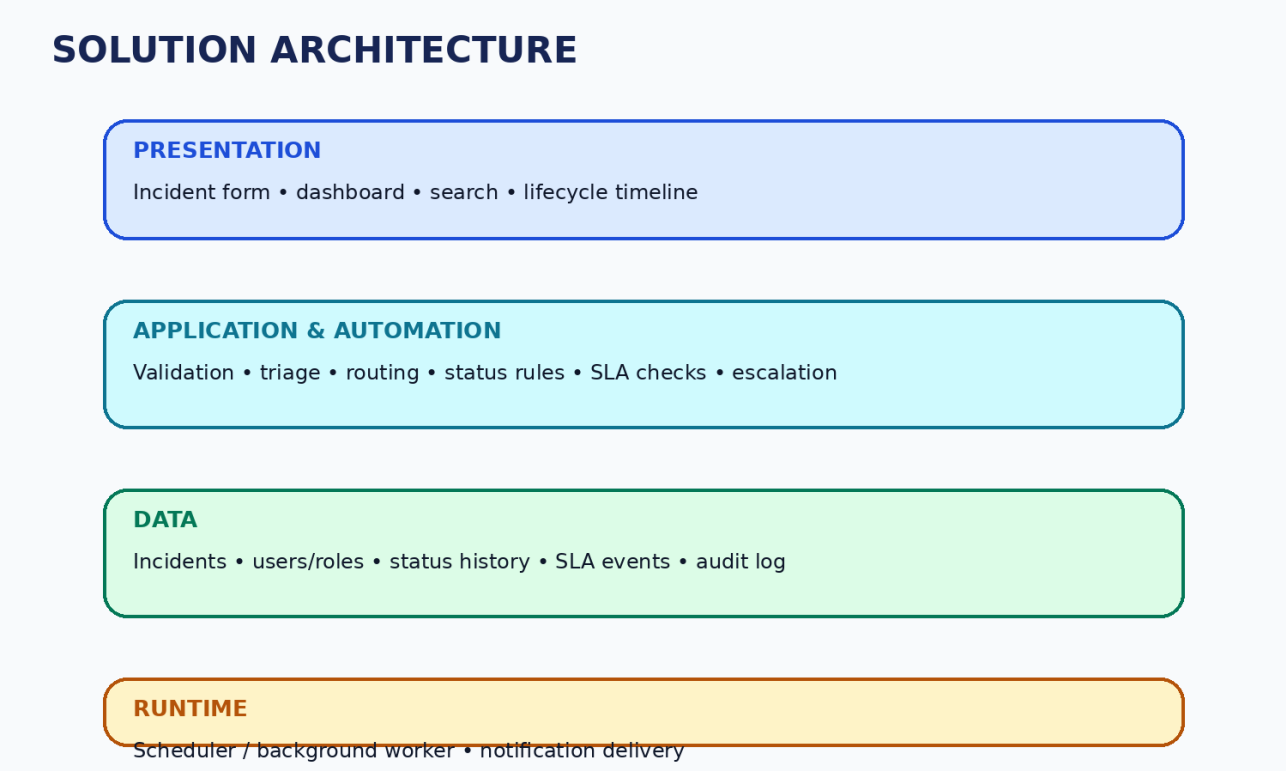
Project Title: Incident Life Cycle Automation
Student Name: Shanmugapriya M

## 2. Project Overview

The system manages operational incidents as a continuous workflow from intake to closure. The design focuses on visibility, ownership, SLA control and traceable history.

- Incident intake and unique identification
- Classification and priority
- Assignment and reassignment
- SLA monitoring and escalation
- Lifecycle status and work notes
- Resolution and closure
- Search, dashboard and audit history

# 3. Architecture




# 4. Logical Data Model

| Entity | Key Data | Relationship |
|---|---|---|
| Incident | ID, title, description, source, category, impact, urgency, priority, status, owner, timestamps | Central record linked to users, history, SLA events and resolution. |
| User / Role | User ID, name, role, access scope | Controls who can view or modify incidents. |
| Status History | Incident ID, old/new status, actor, timestamp, note | One-to-many relationship with Incident. |
| SLA Event | Incident ID, due time, event type, triggered time, state | Tracks SLA and escalation lifecycle. |
| Audit Entry | Actor, action, entity, timestamp, context | Provides traceability for key changes. |

# 5. Service Responsibilities

| Service | Responsibility |
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| Incident Service | Create, retrieve, update and close incidents. |

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| Triage Service | Apply classification and priority rules. |
| Assignment Service | Resolve queue/owner and handle reassignment. |
| SLA Worker | Check due times and trigger escalation events. |
| Notification Service | Deliver assignment, status and SLA messages. |
| Audit Service | Store lifecycle and administrative history. |

## 6. Testing

Functional, UAT and performance testing should be performed against the actual implementation. Results should be recorded from execution rather than assumed.

## 7. Known Limitations

The initial design focuses only on core incident lifecycle automation. Advanced integrations and analytics are not required for the baseline project.

## 8. Future Enhancements

Additional service-specific SLA policies, deeper reporting and selected external integrations can be considered after the core lifecycle is stable.